SpikeShot: Bio-Inspired Coincidence Gating for Neuromorphic Gunshot Detection

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Abstract. Deployed acoustic gunshot detection systems rely on deep convolutional neural networks that are too computationally expensive for battery-powered distributed sensor networks. Spiking neural networks (SNNs) offer a neuromorphically efficient alternative, yet no prior work has applied them to impulsive sound detection. We introduce Coincidence-Gated Temporal Contrast (CGTC) encoding, an audio-to-spike conversion scheme grounded in the physiology of octopus cells in the ventral cochlear nucleus, which fire only when broadband energy changes rapidly and simultaneously across multiple frequency channels. On a 5,044-sample binary dataset of gunshots and acoustically similar urban sounds, three SNN architectures trained on CGTC spike trains all exceed 93% test accuracy with fewer than 48 K parameters, a 400-fold reduction relative to ResNet18. The best-performing model achieves $94.61 \pm 0.47\%$ accuracy (mean over 5 seeds) with 29 K parameters; a separate architecture attains a false positive rate of $1.27 \pm 0.20\%$, roughly $12\times$ lower than that reported for CNN systems evaluated on real-world field recordings. Estimated energy consumption on neuromorphic hardware is 4.5–$5.6\times$ lower than for equivalent artificial neural networks. We further show that event-centric temporal alignment during encoding is critical: accuracy drops by 11 percentage points under a sliding-window protocol.

Keywords: Spiking Neural Networks · Gunshot Detection · Impulsive Audio · Octopus Cells · Bio-inspired Encoding

1 Introduction

The human auditory system detects and responds to a gunshot within tens of milliseconds via dedicated brainstem circuits that rely on sparse, event-driven computation [1], unlike the continuous, resource-intensive processing of current acoustic detection systems [2]. Firearm-related deaths exceeded 43,000 in the United States in 2022 [3], and acoustic sensor networks are now deployed in cities worldwide [5]. Yet commercial platforms such as ShotSpotter remain costly ($65,000 per square mile per year), suffer from alert-fatigue-inducing false positives, and still require human operators for final classification [6]. These limitations follow in part from the deep learning pipelines that underpin such systems.

CNN-based approaches using ResNet18 or multi-CNN ensembles report accuracies above 97% in controlled laboratory conditions [4,7], but rely on models with millions of parameters ill-suited to always-on, battery-constrained edge deployment. The laboratory-to-field gap is severe: the only published CNN evaluation on real-world field recordings achieved approximately 90% accuracy with a 15% false positive rate [27], and few studies report false positive and false negative rates separately despite their importance in safety-critical settings.

Spiking neural networks (SNNs) offer a fundamentally different paradigm that maps naturally onto neuromorphic hardware such as Intel Loihi and IBM TrueNorth [11,12], where energy consumption can be orders of magnitude lower than on conventional processors [9,10]. SNN audio research has so far been confined to speech recognition [13,14] and environmental sound classification [15]; no published work has applied SNNs to impulsive sound detection.

Octopus cells in the ventral cochlear nucleus offer a principled biological model for this task. These neurons are specialised coincidence detectors tuned for broadband transients [16], firing only when the membrane depolarisation rate exceeds 5–15 mV/ms [17] and when multiple tonotopic channels activate within approximately 1 ms [18]. This two-stage filter suppresses both slowly varying backgrounds and narrowband transients, while sub-200 μs spike-timing precision and dendritic compensation for cochlear travelling-wave delays [21] suit them to the rapid, broadband energy signature of gunshots. We translate both properties into Coincidence-Gated Temporal Contrast (CGTC) encoding, a unified audio-to-spike pipeline, and present the first application of SNNs to gunshot detection.

We make three contributions:

1. A bio-inspired CGTC encoding that combines temporal contrast (octopus cell $\mathrm{d}P/\mathrm{d}t$ sensitivity) with broadband coincidence gating in a single pipeline optimised for impulsive sounds.
2. The first SNN-based gunshot detection system, reaching 94.61% accuracy and a false positive rate of $1.27 \pm 0.20\%$ with 29–48K parameters, against published CNN baselines that require two orders of magnitude more parameters, though differences in datasets and evaluation setups limit direct comparison (Section 4.4).
3. A demonstration that event-centric encoding outperforms sliding-window decoding by 11 percentage points, revealing a deployment constraint specific to SNN audio pipelines: any practical system must either supply approximate event timing through a first-stage detector or apply temporal augmentation during training (Section 4.2).

2 Related Work

CNN-based gunshot detectors using ResNet18 [4] or multi-CNN ensembles [7] report accuracies above 97% under laboratory conditions, but field performance is substantially worse: Katsis et al. [27] measured roughly 90% accuracy with a 15% false positive rate in tropical forests, and Papadimitriou et al. [8] showed that CNN recognition degrades at lower signal-to-noise ratios. Few studies report

false positive and false negative rates separately, making deployment readiness difficult to assess.

SNN audio research has focused on speech recognition [13,14] and environmental sound classification [15]. Existing spike encoding strategies trade off temporal and spectral fidelity: rate coding preserves intensity but discards timing [24], time-to-first-spike coding captures onsets but not complex spectra [25], and spectro-temporal approaches [26] lack grounding in specific acoustic phenomena. No prior work has applied SNNs to impulsive sound detection.

Octopus cells in the ventral cochlear nucleus are coincidence detectors requiring synchronised input across tonotopic channels within approximately 1 ms [16], with a low input resistance (7–10 MΩ) that prevents single-band triggering [18]. They fire only when the depolarisation rate exceeds 5–15 mV/ms [17], exhibit sub-200 μs spike jitter [19,20], and compensate for cochlear delays via dendritic mechanisms [21]. These properties make them a natural model for detecting broadband transients such as gunshots.

3 Methodology

3.1 Dataset and Preprocessing

The positive class comprises 2,522 gunshot recordings: 2,148 from the Multi-firearm edge-collected dataset [23], which provides precise timestamp annotations, and 374 from the `gun_shot` class of UrbanSound8K [22]. For the negative class, we selected eight UrbanSound8K categories chosen for acoustic similarity to gunshots (car horn, dog bark, jackhammer, drilling, siren, children playing, engine idling, and street music), sampling 2,522 files to produce a balanced 5,044-sample dataset on which simple energy or spectral features would perform poorly. Files were split at the file level using stratified sampling into training (3,530), validation (757), and test (757) partitions.

All recordings were resampled to 22.05 kHz, converted to mono, and peak-normalised to float32. Each training clip spans 0.75 s, extracted with an asymmetric window of 0.20 s before and 0.55 s after event onset to capture both the initial transient and subsequent reverberation; we retained validation and test recordings at their full 4.0 s duration to support sliding-window evaluation (Section 3.3). Full preprocessing parameters are listed in Table 1.

Multi-firearm timestamp annotations enabled precise event-centric extraction, expanding 1,765 gunshot files to 4,677 clips through multiple shots per file. UrbanSound8K gunshots without annotations were aligned using librosa RMS-based onset detection, selecting the peak via $\max(\Delta\text{RMS}[\text{onset_frames}])$; non-gunshot clips were centred on their energy peak. To restore class balance after extraction, 2,912 additional non-gunshot clips were drawn from files absent in the validation and test splits, yielding a final training set of 9,354 samples (4,677 gunshot, 4,677 non-gunshot) with exact class balance and no file-level leakage.

Table 1: Audio preprocessing parameters.

Parameter	Value
Sample rate	22,050 Hz
Clip duration	0.75 s (16,537 samples)
PRE_OFFSET (before event)	0.20 s
POST_OFFSET (after event)	0.55 s
Onset detection (RMS)	frame=512, hop=128
Energy-peak detection	frame=551, hop=275
Frozen val/test duration	4.0 s (88,200 samples)

3.2 Coincidence-Gated Temporal Contrast Encoding

CGTC encoding converts a raw audio waveform into a one-dimensional binary spike train through four successive operations, each motivated by a specific property of octopus cells in the ventral cochlear nucleus. The pipeline is illustrated in Fig. 1.

The first stage, spectral decomposition, mirrors the frequency analysis performed by the cochlea. We compute a 128-band mel-frequency spectrogram (50 Hz–11 kHz, $n_{\text{fft}} = 1024$, hop = 64 samples) and apply a logarithmic compression $\mathbf{M}_{\log} = \log(\mathbf{M}+\epsilon)$ with $\epsilon = 10^{-6}$. The hop length of 64 samples corresponds to roughly 2.9 ms at 22.05 kHz, comparable to the integration window reported for octopus cells [16].

The second stage, temporal contrast, captures how rapidly the spectral energy is changing rather than its absolute level. For each frequency band f and time frame t we compute the frame-to-frame difference of the log-mel spectrogram:

$$\Delta\mathbf{M}[f,t] = \mathbf{M}_{\log}[f,t] - \mathbf{M}_{\log}[f,t-1] \tag{1}$$

This operation mirrors the dP/dt sensitivity of octopus cells [17]: slowly varying backgrounds produce near-zero differences and are suppressed, while rapid transients such as gunshots yield large positive values. Each frame that exceeds an adaptive threshold $\theta = \mu(\Delta\mathbf{M}) + \sigma(\Delta\mathbf{M})$, computed globally across all $F \times T$ elements, is marked as active.

The third stage, coincidence gating, implements the core octopus cell mechanism. A single output spike is generated at time t only when a sufficient fraction of frequency bands are simultaneously active:

$$s[t] = \mathbf{1}\left[\frac{1}{F}\sum_{f=1}^{F}\mathbf{1}[\Delta\mathbf{M}[f,t] > \theta] \;\geq\; \tau_c\right] \tag{2}$$

We set the coincidence threshold $\tau_c = 0.48$, selected via grid search over $\{0.40, 0.44, 0.48, 0.52, 0.55\}$ (Section 4.3), meaning that at least 48% of the 128 frequency bands must show simultaneous above-threshold activity for a spike to be emitted. The threshold is consistent with physiological observations that octopus cells require substantial, though not complete, convergence of tonotopic input [16,18].

The result of this four-stage pipeline is a one-dimensional binary spike train of approximately 258 timesteps per 0.75-second clip, achieving a 128-fold dimensionality reduction from the original 128-band spectrogram while preserving the temporal structure needed for classification. Fig. 2 illustrates the distinct spike patterns produced for gunshot and non-gunshot sounds: gunshots yield dense, concentrated bursts aligned with the acoustic transient, whereas non-gunshot sounds produce sparser and more scattered activity. Roughly 70% of non-gunshot training clips produce zero spikes under this gate; the remainder exhibit sparse, scattered activity concentrated in broadband transient categories such as jackhammer and dog bark.

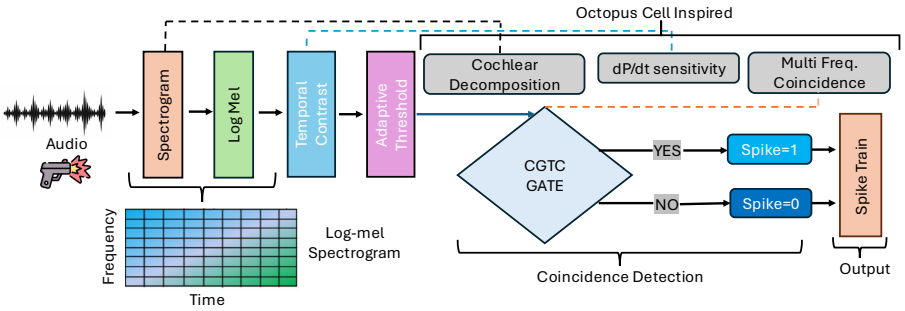


Fig. 1: CGTC encoding pipeline. A log-mel spectrogram is processed through temporal contrast and adaptive thresholding before a coincidence gate emits a spike only when sufficient frequency bands are simultaneously active.

3.3 SNN Architectures and Training

We evaluated three architectures, all using leaky integrate-and-fire (LIF) neurons with membrane decay $\beta = 0.85$ and surrogate gradient learning [28], and all segmenting the input spike train into overlapping windows (size 258, hop 129) before attention-weighted aggregation. HierarchicalSNN (29K parameters) encodes each window through a two-layer LIF encoder (Linear $\rightarrow$ BatchNorm $\rightarrow$ LIF $\rightarrow$ Linear $\rightarrow$ LIF) and combines features via softmax attention. MILSNN (29K parameters) shares the same encoder but adopts a multiple instance learning formulation, producing per-window gunshot probabilities and aggregating at the clip level to directly suppress false positives. TemporalConvSNN (48K parameters) replaces the LIF encoder with parallel multi-scale 1D convolutions (kernel sizes 7, 15, and 31) to capture transient attack and decay at different temporal resolutions, followed by attention pooling.

All models were trained with AdamW (lr $= 0.001$, weight decay 10^{-4}), cosine annealing over 50 epochs, and focal loss [29] ($\alpha = 0.25$, $\gamma = 2.0$), with dropout of 0.3 in attention and classifier modules. The checkpoint maximising validation

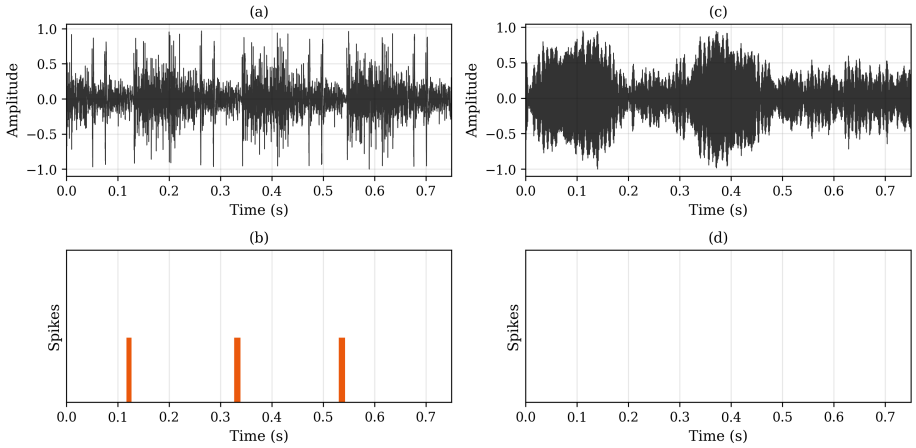


Fig. 2: CGTC spike rasters and corresponding waveforms for a gunshot (a,b) and a non-gunshot urban sound (c,d). The gunshot produces a dense, temporally concentrated burst whereas the non-gunshot yields no spikes. Panel (d) shows one illustrative example only; at $\tau_c = 0.48$, approximately 30% of non-gunshot clips produce sparse non-zero CGTC activity, with transient categories such as jackhammer (47.1%), car horn (24.9%), and siren (19.6%) most likely to spike — the primary source of encoding-level false positives.

F1 was selected. Each model was trained with five random seeds (42–46), with results reported as mean ± standard deviation. Models were evaluated under two protocols: event-centric encoding, where test clips are extracted using timestamp annotations or energy peaks identical to training, and sliding-window evaluation, where the full 4.0 s recording is processed with overlapping 0.75 s windows (hop 0.1875 s) and the maximum window probability is taken as the clip-level decision.

4 Results

We evaluated all three architectures on the held-out 757-sample test set under the event-centric protocol, repeating each experiment across five random seeds. Table 2 summarises the results.

All three architectures exceed 93% accuracy, confirming that CGTC encoding preserves sufficient discriminative information for reliable classification. MIL-SNN achieves the highest accuracy (94.61%) and the lowest false negative rate (7.94%), favouring detection sensitivity. TemporalConvSNN yields the lowest false positive rate (1.27%) and the highest precision (98.59%), reducing alert fatigue at the cost of slightly more missed detections. HierarchicalSNN falls between the two. Standard deviations across the five seeds remain below 1% for accuracy and below 0.01 for F1 and ROC-AUC, indicating stable training.

Table 2: Test set performance with event-centric encoding (n=757). Each cell shows the mean over 5 seeds; standard deviations are below 1% for accuracy and below 0.01 for F1 and ROC-AUC in all cases. Best result per column in **bold**.

Model	Acc. (%)	F1	AUC	FPR (%)	FNR (%)	Params
MILSNN	**94.61**	**0.945**	0.974	2.85	**7.94**	29,380
HierarchicalSNN	94.00	0.937	0.974	1.74	10.26	29,380
TemporalConvSNN	93.79	0.935	**0.976**	**1.27**	11.16	47,843

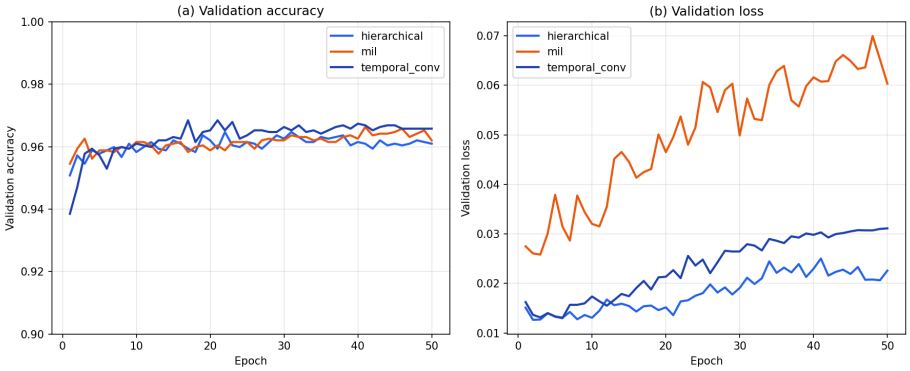


Fig. 3: Training and validation curves over 50 epochs for all three SNN architectures. (a) Accuracy (y-axis: 0.90–1.00): all models converge above 0.93, with MILSNN reaching the highest validation accuracy. (b) Loss: focal loss decreases steadily and plateaus after approximately 30 epochs, confirming stable convergence across architectures.

Fig. 3 shows the learning dynamics of all three architectures. Accuracy stabilises above 0.93 after approximately 30 epochs, and the focal loss trajectories plateau in the same range, confirming that the models converge reliably across all five seeds without signs of overfitting or instability.

4.1 Confusion Matrix and ROC Analysis

To understand how each architecture distributes its errors between false alarms and missed detections, we examined the per-class predictions on the test set at seed 42. The three models reveal a clear accuracy–sensitivity trade-off: MILSNN correctly identifies 347 of 378 gunshots and 372 of 379 non-gunshots, producing 7 false positives and 31 false negatives, giving it the best overall balance. HierarchicalSNN and TemporalConvSNN each favour specificity, producing only 6 and 4 false positives respectively, but missing 33 and 43 gunshots. Fig. 4 visualises these distributions.

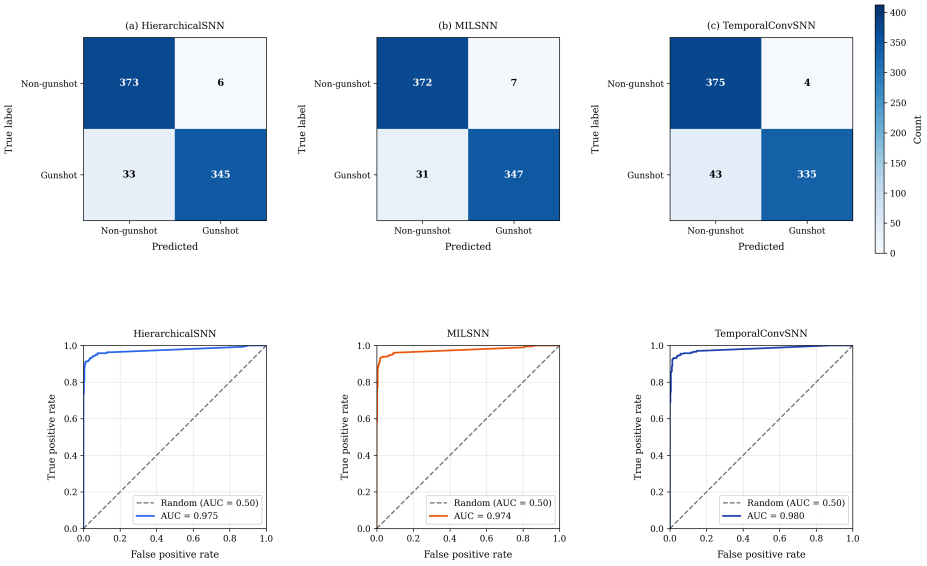


Fig. 4: Confusion matrices (top) and ROC curves (bottom) for all three architectures on the test set ($n = 757$, seed 42). MILSNN achieves the best detection balance, TemporalConvSNN minimises false positives at the cost of more missed detections, and all models achieve AUC ≥ 0.974.

Receiver operating characteristic curves provide a threshold-independent view of discriminative power. All three architectures achieve AUC values between 0.974 and 0.976 (Fig. 4), confirming strong separation across the full range of operating thresholds. TemporalConvSNN reaches the highest AUC (0.976) despite its lower point accuracy, suggesting that its learned representations generalise well when the decision boundary is allowed to shift.

4.2 Event-Centric vs Sliding-Window Encoding

The temporal position of the encoding window has a substantial effect on classification performance. Under event-centric encoding (windows centred on timestamp annotations or energy peaks), MILSNN achieves 94.61% accuracy averaged over five seeds (Table 2). Replacing this alignment with a sliding-window protocol over the full 4-second recording and aggregating via maximum probability drops accuracy to 83.09%, an 11.63 percentage point gap. Averaging the top three window probabilities instead of taking the maximum reduces accuracy further to 77.33% (Table 3). These results show that SNNs are substantially more sensitive to temporal alignment of the input than ANN-based spectrogram classifiers, and that any deployment pipeline must either provide approximate event timing through a first-stage detector or incorporate temporal augmentation during training to reduce this sensitivity.

Table 3: CGTC τ_c sensitivity (MILSNN trained at $\tau_c = 0.48$; test-set encoding only). Bold marks the selected operating point.

τ_c	Acc. (%)	FPR (%)	FNR (%)	Spike Rate
0.10	80.98	16.36	21.69	0.255
0.20	89.04	12.40	9.52	0.058
0.30	92.34	7.65	7.67	0.036
0.40	94.45	3.17	7.94	0.026
0.44	94.72	2.64	7.94	0.024
0.48	**94.98**	**1.85**	**8.20**	**0.022**
0.52	95.24	1.58	7.94	0.020
0.55	94.58	2.11	8.73	0.018
0.60	94.58	0.79	10.05	0.017
0.70	92.73	0.53	14.02	0.013
0.80	89.30	0.26	21.16	0.010

4.3 Parameter Sensitivity and Computational Efficiency

Table 3 extends the τ_c sweep to $[0.10, 0.80]$ with the MILSNN classifier fixed at $\tau_c = 0.48$. Accuracy stays within 0.8 percentage points across $\tau_c \in [0.40, 0.60]$, confirming a flat stable region. Below 0.30, rising spike rate drives FPR above 7.6%; above 0.70, spike collapse pushes FNR above 14%. The selected value $\tau_c = 0.48$ was chosen by validation F1 and sits within this stable region. The threshold multiplier peaked at 1.0; deviations in either direction traded spike rate against discriminability monotonically. All three architectures achieve sub-millisecond inference latency with fewer than 48 K parameters and 187 KB of memory, and estimated energy consumption on neuromorphic hardware is 4.5–5.6× lower than for equivalent ANNs [30,11], though these figures await validation on physical hardware.

4.4 Comparison with Published Methods

Direct comparison across studies is limited by differences in datasets, recording conditions, and evaluation protocols. CNN systems evaluated under controlled laboratory conditions report accuracies near 99% [4,7], but the only field-deployed CNN evaluation achieved approximately 90% accuracy with a 15% false positive rate [27]. Our TemporalConvSNN attains a substantially lower false positive rate (1.27%) with two orders of magnitude fewer parameters, and all three SNN architectures exceed 93% test accuracy with stable performance across five random seeds (Table 4).

5 Discussion and Future Work

The effectiveness of CGTC encoding stems from two properties inherited from octopus cell physiology. Temporal contrast suppresses slowly varying backgrounds

Table 4: Comparison with published methods and our validation results. †Controlled datasets. Our results (below the mid-rule) report the mean over 5 seeds; standard deviations are below 1% for accuracy in all cases.

		Test		Validation		
Method	Type	Acc. (%)	FPR (%)	Acc. (%)	FPR (%)	Params
ResNet18 [4]	CNN	99.0†	N/A	–	–	11.7M
CNN Ensemble [7]	CNN	99.0†	N/A	–	–	∼1M
CNN (Field) [27]	CNN	∼90	15.0	–	–	∼5M
MILSNN (ours)	SNN	94.61	2.85	93.53	4.60	29K
HierarchicalSNN (ours)	SNN	94.00	1.74	92.95	3.17	29K
TemporalConvSNN (ours)	SNN	93.79	1.27	93.58	2.01	48K

while preserving rapid amplitude excursions characteristic of gunshots, and coincidence gating rejects narrowband transients that activate only a subset of frequency bands. Together, these stages produce spike trains that are dense and concentrated for gunshots but absent for acoustically similar urban sounds (Fig. 2), providing the downstream SNN with a clean discriminative signal.

The 11 percentage point accuracy gap between event-centric and sliding-window encoding reveals that SNNs are considerably more sensitive to temporal alignment than CNN-based spectrogram classifiers. Any deployment pipeline must therefore either incorporate a lightweight onset detector as a first stage or apply temporal augmentation during training; this finding likely generalises beyond gunshot detection to SNN-based audio classification more broadly. A natural realisation would pair a low-power energy-threshold detector running on a microcontroller with the CGTC-SNN classifier, which operates only on candidate segments to maintain low average power consumption.

Several limitations warrant further investigation. The estimated 4.5–5.6× energy advantage over ANNs has not been validated on physical neuromorphic hardware such as Intel Loihi, where data movement and idle-state power may shift these ratios. The global adaptive threshold in CGTC encoding could be extended to per-band or per-frame adaptation to improve robustness in non-stationary environments. Finally, validation on larger and more diverse corpora, and expansion of the negative class to include fireworks, construction blasts, and vehicle backfires, would provide stronger evidence of generalisation to real-world deployment conditions.

6 Conclusion

We have presented the first application of spiking neural networks to gunshot detection, introducing Coincidence-Gated Temporal Contrast encoding, an audio-to-spike conversion scheme whose temporal contrast and coincidence gating stages are grounded in the physiology of octopus cells in the mammalian

cochlear nucleus. On a 5,044-sample binary dataset of gunshots and acoustically similar urban sounds, all three SNN architectures achieve competitive accuracy and false positive rates with parameter counts two orders of magnitude below those of published CNN baselines. The highest accuracy (94.61%) is obtained by MILSNN with 29 K parameters, while TemporalConvSNN achieves the lowest false positive rate (1.27%) with 48 K parameters, substantially below the 15% reported for a field-deployed CNN, though differences in datasets and conditions limit direct comparison. The sensitivity of SNN classifiers to temporal alignment, quantified here as an 11 percentage point accuracy gap between event-centric and sliding-window protocols, constitutes a practical constraint that distinguishes spike-based from spectrogram-based audio processing and should inform the design of any deployment pipeline. Estimated energy consumption on neuromorphic hardware is 4.5–5.6 times lower than for equivalent artificial neural networks, and if confirmed through deployment on platforms such as Intel Loihi, this advantage would bring always-on, battery-powered acoustic monitoring within practical reach.

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